

Root Test of Factorial and Beyond

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Abstract

This article extends the regular college level calculus II into a higher level by putting root test in a series with factorial, and furthermore study about the n th root of factorial ($\sqrt[n]{n!}$). By studying this specific limit ($\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n}$), we found that there is some connections between the n th root of factorial and The Optimal Stopping Problem. Also, by comparing the root test and ratio test, we've also found some interesting connections between those.

1 Introduction

First, I want to introduce the question that inspired me:

$$\sum_{n=1}^{\infty} \frac{n!}{n^n} \quad (1)$$

And we want to show whether this is convergent. Regularly, we can use the Ratio Test, because obviously there is a factorial.

$$\begin{aligned} L_{\text{Ratio}} &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1) \cdot n! \cdot n^n}{(n+1) \cdot (n+1)^n \cdot n!} \\ &= \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n} \\ &= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n \\ &= \lim_{n \rightarrow \infty} \left(\frac{n+1-1}{n+1} \right)^n \\ &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)^n \end{aligned} \quad (2)$$

From here, we can see that the limit is similar to the definition of e , and this types of limit is actually e^a . The reader may easily prove the limit is $1/e$.

$$L_{\text{Ratio}} = \lim_{n \rightarrow \infty} \left(1 + \frac{-1}{n+1} \right)^n = e^{-1} = \frac{1}{e} < 1 \quad (3)$$

So we've showed that by Ratio Test, the series converges. But this is not fun, because it's just a regular calculus II question. Now, let's try using root test to do the same thing, because there is a n^n .

$$\begin{aligned}
 L_{\text{Root}} &= \lim_{n \rightarrow \infty} \sqrt[n]{a_n} \\
 &= \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n!}{n^n}} \\
 &= \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n}
 \end{aligned} \tag{4}$$

This limit is very important, and our topic is going to extend by this limit. Now, the problem here is the n th root of factorial can't be simplified easily. So I am going to use the numerical method to solve it. If you want to see the proof, please go Appendix A, the proof was really beautiful.

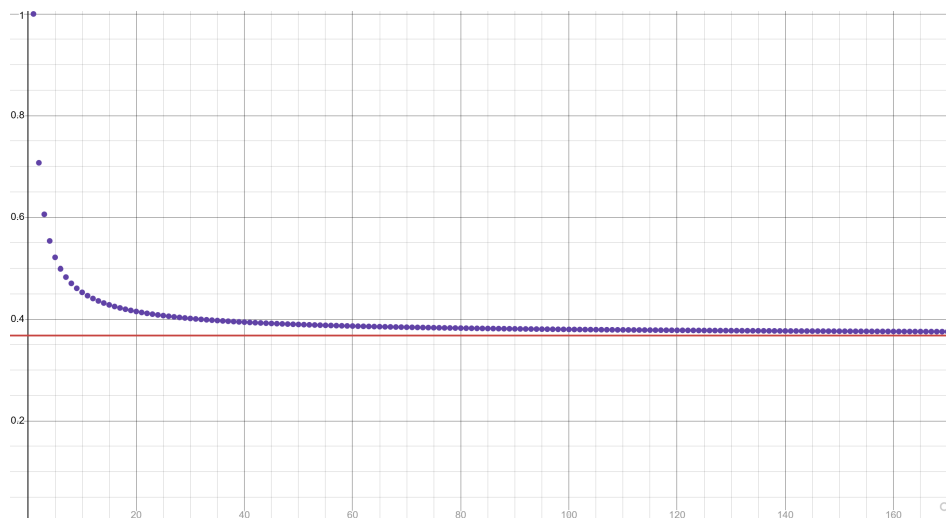


Figure 1: Numerical Verification of the Limit in (4)

I used a calculator to graph the first 170 points of the equation (the highest the calculator can get), and having a guess of the limit approaches $1/e$. This reasonable guessing was taken from the solution in Ratio Test. This led to my 2 conjectures, and it's natural to extend:

1. How does solution $1/e$ connect to The Optimal Stopping Problem?
2. If the limit exists, how do the limit of Root Test and Ratio Test have the same behavior?

2 The Optimal Stopping Problem and $1/e$

2.1 The Optimal Stopping Problem

When I got the result, I was shocked about the solution, because $1/e \approx 37\%$ occurred in a YouTube video made by the YouTuber Veritasium, which specifically talked about The Optimal Stopping Problem [2].

To illustrate this, we introduce the classical formulation of the Secretary Problem, historically also framed in literature as the *Marriage Problem* (see, e.g., Ferguson [1]). Note that this specific setting is adopted solely for clarity and ease of understanding, without any intended bias or targeting. Consider a man who wants to get married within 10 years (from the age of 22 to 32), and he wants to pick the best girl he meets in these ten years. However, he can't go back and pick a girl he met before, meaning he needs to decide whether he is going to marry the girl right at the time or pass. We now want to know what the best strategy is for him to have the highest probability of marrying the best girl.

Let $t \in [22, 32]$ be the time in years. We want to find a stopping point $t_S \in [22, 32]$ to use the following strategy:

- When $22 \leq t \leq t_S$, the man only observes the girls to see how good they are. He does not marry anyone in this period.
- When $t_S < t \leq 32$, he starts looking for a wife. If he meets a girl who is better than all the girls he met before t_S , he marries her immediately.



Figure 2: Timeline of the optimal stopping strategy from age 22 to 32.

So, our best strategy turns out to be: What t_S can maximize the probability of him to marry the best girl?

Let's first try some extreme value, that $t_S = 22$ and $t_S = 32$.

- $t_S = 22$, he starts to marry at the age of 22, meaning he will 100% marry the first girl, because she is the best after S.
- $t_S = 32$, he is never going to marry until the age of 32. He is forced to pick the last girl and marry.

Obviously those 2 t_S values are the worst, because they are literally equivalent to choose any girl randomly and just to marry.

To start the proof, we let N be the total number of girls he met in 10 years. We assume they appear at a constant rate, so time $t \in [22, 32]$ maps to number i girl by the linear relation:

$$i(t) = \frac{N}{10}(t - 22)$$

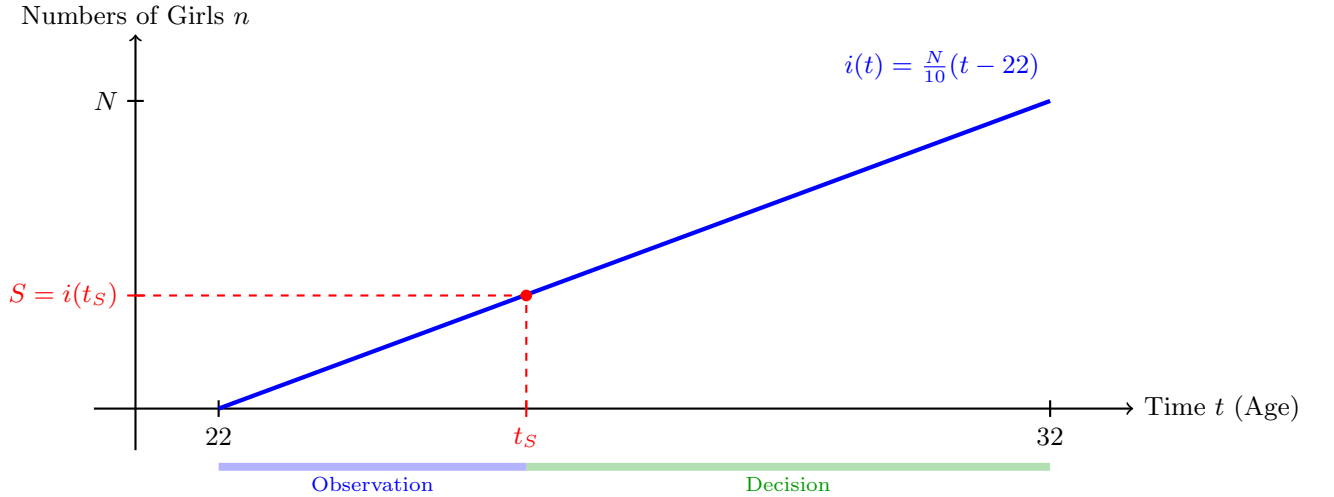


Figure 3: Linear mapping between time t and numbers of girls n

Where we define S be the S -th girl he observed, and starts considering to make the decision after S .

The probability of “successfully choose the best option” can be calculated by adding up all the probabilities ($\sum$) of ”successfully choose the n th solution”, $\forall n \in \mathbb{N}, S < n \leq N$, so that we can know the probability of getting the best option. For the number n th girl ($S < n \leq N$), we have:

- the probability of that girl is the best girl overall: $\frac{1}{N}$
- The probability of the strategy successfully chooses the girl: all options k in $S < k \leq n$ is less than the maximum of first S options. Which means we want the best of first $(n - 1)$ options is less than S . So the probability of that happen is $\frac{S}{n-1}$.
- We multiply them, which is the probability of the n th girl is best, and be chosen.

So we add all cases together:

$$P(\text{successfully choose the best option}) = \sum_{n=S+1}^N \frac{1}{N} \cdot \frac{S}{n-1} \quad (5)$$

$$= \frac{S}{N} \sum_{n=S+1}^N \frac{1}{n-1} \quad (6)$$

Notice that (6) is true because S and N are both constant respect to the summation. Now, we can change the lower limit and upper limit by substitute $i = n - 1$, where $n = i + 1$, to rewrite the summation prettier.

$$P(\text{successfully choose the best option}) = \frac{S}{N} \sum_{i+1=S+1}^{i+1=N} \frac{1}{i} = \frac{S}{N} \sum_{i=S}^{N-1} \frac{1}{i} \quad (7)$$

To deal with the harmonic series easily, we can let $N \rightarrow \infty$, and change the series into integral, so we can use the analysis methods.

$$x_i := \frac{i}{N}, \Delta x := \frac{1}{N}$$

$$\begin{aligned} \lim_{N \rightarrow \infty} P(\text{successfully choose the best option}) &= \lim_{N \rightarrow \infty} \frac{S}{N} \sum_{i=S}^{N-1} \frac{1}{i} \\ &= \lim_{N \rightarrow \infty} \frac{S}{N} \sum_{i=S}^{N-1} \frac{1}{\frac{i}{N} \cdot N} \\ &= \lim_{N \rightarrow \infty} \frac{S}{N} \sum_{i=S}^{N-1} \frac{1}{i/N} \cdot \frac{1}{N} \\ &= \lim_{N \rightarrow \infty} \frac{S}{N} \sum_{i=S}^{N-1} \frac{1}{x_i} \cdot \Delta x \end{aligned} \quad (8)$$

By considering S/N as x_S , we can rewrite the Riemann Sum into a definite integral. Lower bound: x_S , Upper bound: $\lim_{N \rightarrow \infty} (N - 1)/N = 1$

$$\begin{aligned} (8) &= x_S \int_{x=x_S}^{x=1} \frac{1}{x} \cdot dx \\ &= x \int_x^1 \frac{1}{t} dt \\ &= x [\ln t]_x^1 \\ &= x [\ln 1 - \ln x] \\ &= -x \ln x \end{aligned} \quad (9)$$

This is the function of x where x is the ratio of stopping point S and total number N , when N approaches infinity. This is the ultimate probability of choose the best girl.

So, we want to maximize the function, to find the best value of S/N .

$$\begin{aligned} \frac{dy}{dx} = 0 &\iff \frac{dy}{dx} = \frac{d}{dx}(-x \ln x) \\ &= -\ln x + (-1) = 0 \\ \implies -\ln x = 1 &\implies \ln x = -1 \implies x = e^{-1} = 1/e \end{aligned} \tag{10}$$

At the point where the ratio of S and N are $1/e$, there is a highest probability of “successfully choose the best girl”. For example, in this scenario, the best t_S is $22 + 10/e \approx 25.7$ years.

We can notice that this ratio is equal to the limit we discovered before: $\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n}$ (4)

2.2 Connections of $1/e$

Let's rewrite the equations:

$$\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n} = \frac{S}{N} = \frac{1}{e} \tag{11}$$

We want to see what is the connections.

References

- [1] Thomas S. Ferguson. Who solved the secretary problem? *Statistical Science*, 4(3):282–289, 1989.
- [2] Veritasium. Why is this number everywhere? YouTube video, March 2024. Available at <https://www.youtube.com/watch?v=d6iQrh2TK98> (Accessed: 2026-06-03).